AMEYA BHAMARE
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CORE SKILLS

- **GenAI:** LangChain, Prompt Optimization, Hallucination Analysis, CrewAI, RAG, MCP, n8n, Ollama, TTS/STT
- **Evaluation & MLOps:** Model Observability & Governance, CI/CD, MLflow, SHAP, A/B Testing, LLM-as-a-Judge
- **Engineering:** Python, PySpark, SQL, Databricks, AWS, SDLC, Git, Docker, Postman, Cloudflare, FastAPI, pgvector

RELEVANT EXPERIENCE

Senior Data Scientist | Centene | SF Bay Area May '24 - Present

- Architected a **competitive-intelligence chatbot** using **LangGraph** and RAG. Built **multi-agent workflows** for retrieval, reranking, generation, and explanation. Deployed **LLM-as-a-Judge** to assess coverage and grounding. Routed low-confidence outputs to human review, creating flywheels that **raised reviewer throughput by 3x**.
- Deployed Quality Rating System models end-to-end in Databricks **across 30+ markets** replacing Excel workflows. Built **MLflow eval pipelines** (tracing, drift, calibration), cutting manual effort by 95% **saving ~500 hours annually.**
- Led a **5-person team** rebuilding the claims forecasting framework after the incumbent system missed forecasts by millions of dollars. Designed a **novel four-factor PMPM decomposition method** that uses ML exclusively to isolate population-mix effects. **Reduced MAPE from ~33% to within 12% across 6 cost categories and 6.7% bias**.
- Built **internal Python packages** using Poetry, an **AI compliance review system** and an auto-presentation toolkit **saving ~250 hours annually**. **Managed technical delivery and pre-handoff validation** for client deployments.
- Re-architected the IMP1A pipeline in R using httr2. **Reduced E2E runtime from ~480 to 15 minutes (32x)** with concurrent requests, preflight checks, paginated retrieval, retries, throttling, and structured logging.
- Drove org-wide Databricks enablement for **65 analysts**. **Saved 80+ onboarding hours** and **~$10K in projected training costs**. Mentored 2 analysts through E2E ML projects. Pitched the need for an internal AI-assisted dev platform with **PHI/PII guardrails** to IT leaders, secured beta access, and now train my team on agentic workflows.
- Shipped a multi-tier pricing dashboard for **actuarial clients** with **zero production crashes** in its first three months.
- Defined the org's **Python package deployment standards** with the platform team. Auto-published docs using great-docs, CI/CD, git hooks for commit standards, auto-tagged releases with changelogs on Posit Connect, issue templates, and release-branch publishing using **git-builder**.

AI Researcher | Amazon | Seattle Oct '23 - Mar '24

- Designed and evaluated a **SARIMA + Gaussian copula–based generative model** to synthesize S3 storage request workloads, capturing seasonality and cross-feature dependencies for non-disruptive performance experiments.
- Proved the robustness via an MVP on **271M encoded time-series records** achieving **74% distributional similarity** and **98% normalized frequency preservation** relative to production data.

Machine Learning Intern | Centene | St. Louis Jun '23 - Sep '23

- Implemented a **hybrid NLP entity-recognition architecture** combining rule-based extraction and domain-specific biomedical embeddings to parse ICD-10 codes and clinical attributes within medical claims at **~96% accuracy**.
- Leveraged PubMed vector integration and transfer learning to replace AWS Comprehend Medical, **projecting $25-27M in annual savings**.

Previous Roles | Bangalore, India Aug '19 - Mar '22

- **AI Consultant | Ernst & Young:** Improved recommender ranking metrics by 9x with graph-based models and built Azure Data Factory pipelines evaluating 10,000+ model submissions
- **Deep Learning Consultant | Bounce**: Deployed real-time license plate OCR, lifting scooter-rental bookings by 17%

PROJECTS

CareGap Voice | Personal Project | github.com/ameyabhamare/caregap-voice 2026

- Built a **voice AI agent** for health plan care-gap outreach on synthetic FHIR data. Implemented gap detection using PySpark scripts. Gated LLM scripts via **compliance rules** and human review. Used **Kokoro TTS** to create voice transcripts and setup **n8n/Twilio workflows** to make the calls. New plans can be added using zero code changes.
- Built a **RAG callback agent** with a model-free grounding check with **zero spoken answers mentioning a wrong amount across 88 golden questions** and **100% refusal of medical and off-topic questions**.
- Raised script compliance from **89% to 99.7%** with **MLflow prompt evals** and a **CI eval gate**. Benchmarked Claude against a local model reporting a **callback accuracy increase from 79% to 100%** for $0.88.

EDUCATION

- **Data Science (Master of Science) | University of Washington | Seattle** **GPA - 3.9 / 4**
- Computer Science (Bachelor of Technology) | PES University | Bangalore GPA - 3.9 / 4